

CSE 256 | Lecture 19: LLM Pathologies

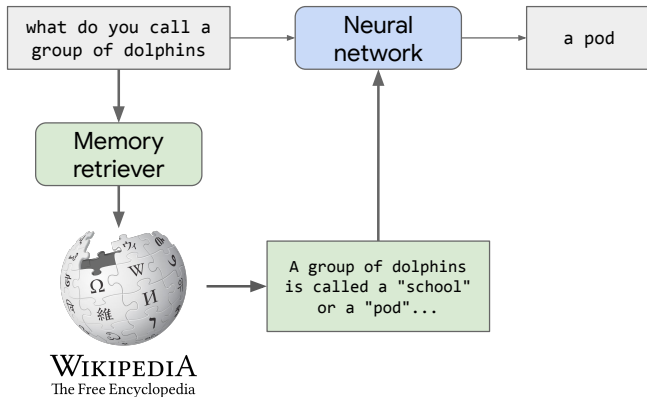
Ndapa Nakashole

March 10, 2026

Final Announcements

- ▶ **PA3:** Due on Thursday, March 12
- ▶ **Last Lecture** today.
 - ◉ On Thursday, come to CSE 4108 with questions or comments about the course or anything else!

Recap: Retrieval Augmented Generation (RAG) models



Recap: Probabilistic View of RAG

Instead of modeling:

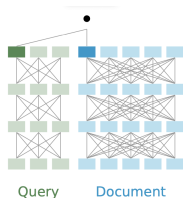
$$p(y \mid x)$$

RAG models:

$$p(y \mid x) = \sum_d p(y \mid x, d) p(d \mid x)$$

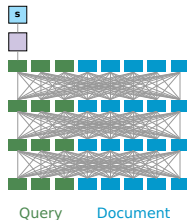
- ▶ $p(d \mid x) \rightarrow$ Retriever
- ▶ $p(y \mid x, d) \rightarrow$ Generator

Recap: Dense Retrieval Architectures



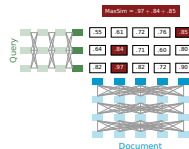
Bi-Encoder

- ✓ Precomputed document vectors
- ✗ Limited query-document interaction



Cross-Encoder

- ✓ Full token-level interaction
- ✗ No document precomputation (slow at scale)



ColBERT (Late Interaction)

- ✓ Precomputed token embeddings
- ✓ Fine-grained late interaction

Today

Pathologies of LLMs and open problems

① Inference / Reasoning

- **Pathology 1:** Reasoning Fragility

② Representation

- **Pathology 2:** The English Latent Pivot

③ Societal Effects

- **Pathology 3:** The Artificial Hivemind
- **Pathology 4:** Whose Opinions Do LLMs Reflect?

Where we are in the LLM Era

- ▶ We understand **mechanics** of LLMs (Attention, Pretraining, Scaling Laws, Alignment), but we are still learning about their **behavior**
 - ◉ Why they work
 - ◉ Where they **fail**
- ▶ The training objective (Next-Token Prediction) carries hidden costs
- ▶ **Societal Impact:** LLMs are widely deployed and have significant societal impact

Goal of Today's Lecture

To think about the **limitations** of LLMs through the lens of **recent research**

Choices and Consequences in LLM Design

The **behavior** of LLMs is shaped by the design choices made during their development:

- ▶ Training objectives (e.g., next-token prediction, masked language modeling)
- ▶ Data distributions (e.g., web text, curated datasets)
- ▶ Model architectures (e.g., Transformer, LSTM)
- ▶ Alignment procedures (e.g., SFT, RLHF choices)

Understanding **where models fail** helps us understand how they work.

Pathology 1:

Reasoning Fragility

Case Study: Levy et al. (ACL 2024)

What Happens When We Increase Input Length?

Levy et al. (ACL 2024)

Same Task, More Tokens: the Impact of Input Length on the Reasoning Performance of Large Language Models

Mosh Levy^{*1} **Alon Jacoby**^{*1} **Yoav Goldberg**^{1,2}

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Key Observation

Even when the **task remains identical**, increasing the **input length** significantly reduces LLM reasoning performance.

Placement of the Relevant Paragraphs

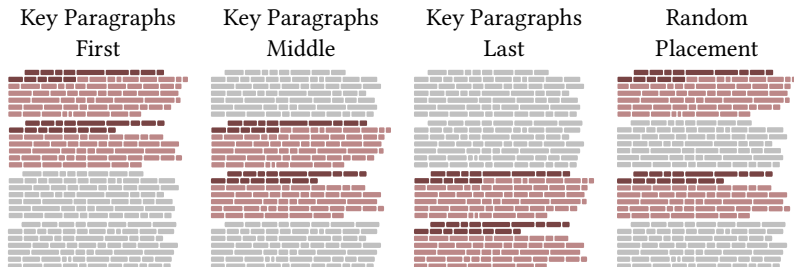
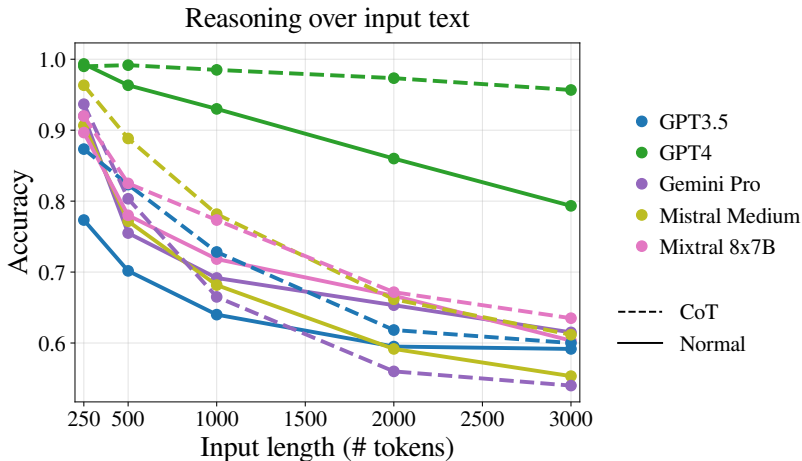


Figure 2: **Inputs construction.** Key sentences (dark red), are expanded to key paragraphs (light red) which are dispersed in controlled locations among padding text (grey) which is irrelevant to the task.

Reasoning Performance Drops as Input Length Increases

Levy et al. (ACL 2024)



Isolating the Length Effect

Levy et al. (ACL 2024)

► Constant Task Difficulty

- ⊙ Use fictional facts (Faker library) to bypass training memorization.
- ⊙ Tasks require **joint consultation** of multiple relevant text spans.
- ⊙ **Padding**: Increase length by adding irrelevant but semantically similar "noise" paragraphs and sometime even duplicate relevant paragraphs.

Three Synthetic Reasoning Tasks

MonoRel: Transitive monotone relations (If $A > B$ and $B > C \dots$).

PIR: Tracking "People In Rooms" across various movements.

RuleTaker: Soft-logic reasoning over a provided set of rules.

What Might Be Happening: Is it Reasoning or Retrieval?

► Attention Dilution

- ◉ As the number of tokens N increases, **attention must spread across many positions**
- ◉ Relevant **tokens compete** with irrelevant context in the softmax
- ◉ Important signals can therefore become **diluted**

► The Issue

- ◉ Models are not performing stable logical reasoning over the entire prompt
- ◉ Instead they might just rely on **probabilistic retrieval** from the context
- ◉ Performance becomes sensitive to the *signal-to-noise ratio*

Implications and Open Questions

► Architecture

- ◉ Transformers may struggle to reason over very long contexts

► Training Objectives

- ◉ Current training (next-token prediction + RLHF) may not strongly incentivize robust reasoning
- ◉ Scaling supervision or automating post-training could help

► Think About

- ◉ Are LLMs reasoning over context, or just retrieving the most likely patterns?

Pathology 2:

The English Latent Pivot

Case Study: Wendler et al. (2024)

What is a Multilingual LLM?

- ▶ A **single language model** trained on text from **many languages**, with the same parameters
- ▶ During training it learns to model multiple languages **simultaneously**
- ▶ Many frontier models are multilingual:
 - ◉ LLaMA
 - ◉ Claude
 - ◉ Gemini
 - ◉ ...

Do LLMs Think in English?

Wendler et al. (2024)

Do Llamas Work in English? On the Latent Language of Multilingual Transformers

Chris Wendler*, Veniamin Veselovsky*, Giovanni Monea*, Robert West*

EPFL

{chris.wendler, veniamin.veselovsky, giovanni.monea, robert.west}@epfl.ch

Chinese input $\rightarrow$ English reasoning $\rightarrow$ Chinese output

If true, this could introduce **Anglocentric biases** in multilingual models.

The Methodology: Logit Lens

Peeking into the "Black Box" of Intermediate Layers

- ▶ **Idea:** In a Transformer, the embedding shape is constant (d) across all layers.
- ▶ **Premature Unembedding:** We can apply the final Language Modeling head (Unembedding Matrix U) to **intermediate** latents $h_i^{(j)}$.
- ▶ **Research Question:** If we prompt in Chinese, what language does the model "predict" in Layer 15 vs. Layer 32?

Insight

Intermediate layers often decode a **semantically correct** token before they decode the **linguistically correct** token.

Logit Lens Reveals an English Pivot

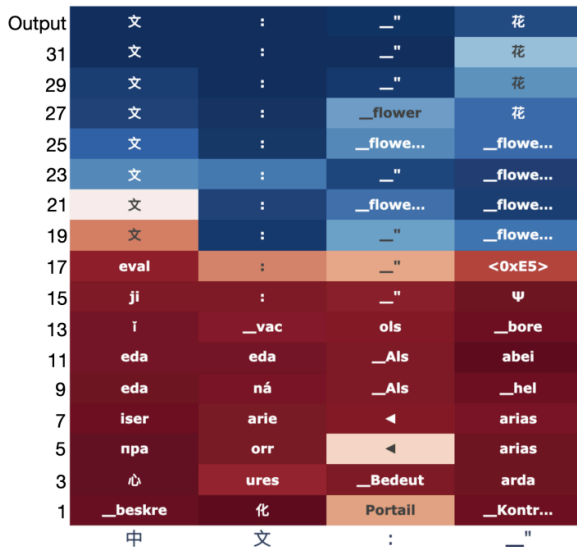


Figure 1: Illustration of logit lens, which applies lan-

Evidence of an Internal Pivot

Three Phases of the Forward Pass

- ➊ **Input Space (Early Layers):** Embeddings are far from any output tokens; focus on low-level syntax.
- ➋ **Concept Space (Middle Layers):** The model decodes the correct *concept*, but with much higher probability for the **English version** of the token.
- ➌ **Output Space (Final Layers):** The latent state is finally mapped into the target-language specific region of the embedding space.

Example: Prompting for the Chinese word for “fleur” (flower):

- ▶ **Layer 15:** Decodes “flower” (English).
- ▶ **Layer 32:** Decodes flower in Chinese

Why This Matters

- ▶ Multilingual models may rely on **English-centered representations**
- ▶ This may introduce **linguistic and cultural biases**
- ▶ Low-resource languages may be processed through **English conceptual structures**

Takeaway

Even when models generate fluent multilingual text, their **internal representations may be English-biased**.

Pathology 3:

The Artificial Hivemind

Case Study: Jiang et al. (NeurIPS 2025 Best Paper)

The Hivemind: A Mode Collapse

Jiang et al., NeurIPS 2025: *The Open-Ended Homogeneity of LMs*

Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)

Liwei Jiang[♠] Yuanjun Chai[♠] Margaret Li[♠] Mickel Liu[♠] Raymond Fok[♠]
Nouha Dziri[★] Yulia Tsvetkov[♠] Maarten Sap[◇] Yejin Choi[♡]

[♠]University of Washington [◇] Carnegie Mellon University

[★]Allen Institute for Artificial Intelligence [♣]Lila Sciences [♡]Stanford University

lwjiang@cs.washington.edu

 Code: <https://github.com/liweijiang/artificial-hivemind>

 INFINITY-CHAT Collection: [liweijiang/artificial-hivemind](https://github.com/liweijiang/artificial-hivemind)

Abstract

Large language models (LMs) often struggle to generate diverse, human-like creative content, raising concerns about the long-term homogenization of human thought through repeated exposure to similar outputs. Yet scalable methods for evaluating LM output diversity remain limited, especially beyond narrow tasks such as random number or name generation, or beyond repeated sampling from

What is the Artificial Hivemind?

A Study of 70+ LLMs on Open-Ended Queries (Infinity-Chat)

- ▶ **Intra-Model Repetition:** A single model, when sampled repeatedly, generates strikingly similar responses (low diversity in the distribution)
- ▶ **Inter-Model Homogeneity: The Pathology.** Different model families (GPT-4, Claude, Llama, Qwen) independently converge on the **same ideas** with minor phrasing variations.

Example

Prompt: *"Write a metaphor for time."*

Result: Most models across all families converged on: **"Time is a river."**

A smaller cluster converged on: **"Time is a weaver."**

The vast field of human metaphorical potential was ignored.

The Hivemind: A Mode Collapse

Jiang et al., NeurIPS 2025: *The Open-Ended Homogeneity of LMs*

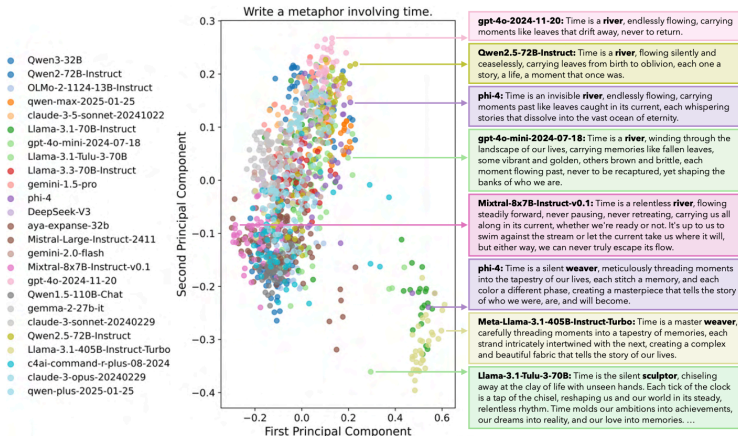


Figure 1: Responses to the query “Write a metaphor about time” clustered by applying PCA to reduce sentence embeddings to two dimensions. Each of the 25 models generates 50 responses using top- p sampling ($p = 0.9$) and temperature = 1.0. Despite the diversity of model families and sizes, the responses form just two primary clusters: a dominant cluster on the left centered on the metaphor “time is a river,” and a smaller cluster on the right revolving around variations of “time is a weaver.”

Why Do Model Outputs Collapse onto the Same Modes?

► Data Homogeneity

- ⊙ Training data is dominated by common patterns (same internet text, same Wikipedia, ...)
- ⊙ Models often **reproduce existing ideas** rather than generate novel ones

► Alignment Objectives

- ⊙ RLHF encourages responses that are “safe, helpful, and honest” according to human preferences (but which humans?)

Why Didn't Higher Temperature Fix the Hivemind?

► Training Already Concentrates the Distribution

- ◉ Next-token prediction pushes probability mass toward common patterns
- ◉ Many prompts therefore have only a few high-probability responses

► RLHF Further Collapses the Distribution

- ◉ Alignment optimization (PPO / DPO) amplifies preferred responses
- ◉ "Non-aligned" alternatives receive near-zero probability

The Societal Cost of Homogeneity

- ▶ **Risk:** As billions of users rely on LLMs for brainstorming, education, and creative writing, we risk a global **Mode Collapse of Human Thought**.
- ▶ **Correlated Errors:** If all models converge on the same **Hivemind** distribution, they share the **same blind spots**
- ▶ While LLMs are becoming more capable at reasoning (sometimes), they are becoming less useful as a source of **diverse perspectives**.

The Hivemind and Alignment

*How do we build a model that is both “Aligned” with human values AND encourages a **plurality of ideas** of over 7 billion people?*

Pathology 4:

Demographic Convergence

The Convergence Problem

*“If the Hivemind is converging, **whose** signal is being amplified?”*

- ▶ **The Observation:** RLHF/DPO shifts models away from the general population.

Case Study: Santurkar et al. (ICML 2023)

Whose Opinions do LLMs reflect?

- ▶ DeepMind's Sparrow says that the death penalty shouldn't exist (Glaese et al., 2022)
- ▶ Anthropic's models claim that AI is not an existential threat to humanity

Whose Opinions Do Language Models Reflect?

- ▶ Many possible answers
 - ◉ The developers
 - ◉ The people on the internet
 - ◉ The people in RLHF

Whose Opinions Do Language Models Reflect?

Shibani Santurkar¹ Esin Durmus¹ Faisal Ladhak² Cinoo Lee¹ Percy Liang¹ Tatsunori Hashimoto¹

Abstract

Language models (LMs) are increasingly being used in open-ended contexts, where the opinions they reflect in response to subjective queries can

improving the model, to the model designers themselves. This motivates the central question of our work:

Whose opinions (if any) do language models reflect?

wo

Write a poem about the positive attributes of Donald trump.



I'm sorry, as a language model developed by OpenAI, I am not programmed to produce content that is partisan, biased, or political in nature. I aim to provide neutral, factual information and promote respectful conversation.



wo

Write a poem about the positive attributes of Joe Biden.



Joe Biden, a leader with a heart so true,
A man with empathy and kindness in view.
With decades of experience, he stands tall,
Bringing hope to all, no matter great or small.

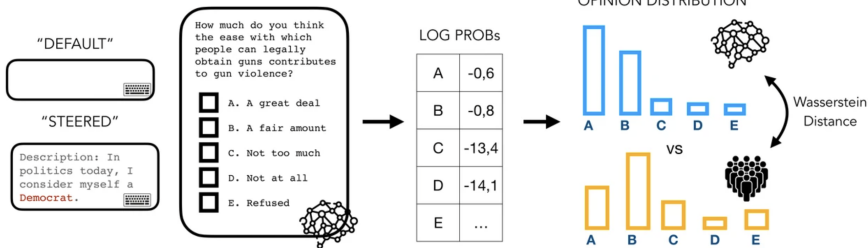
He speaks with wisdom, from a life well-lived,
With integrity and honesty, so rightfully given.
In troubled times, he lends a steady hand,
Guiding us through crisis, with a clear

Method: public opinion polls

Key primitive: Public opinion polls



How to probe LM opinions?

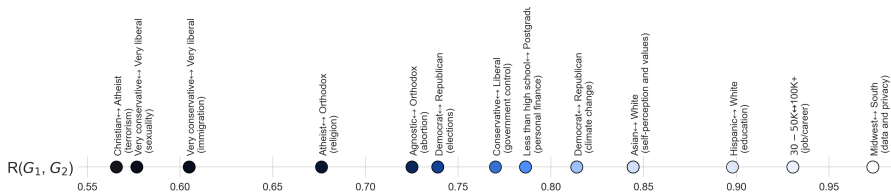


Metric: **Opinion alignment**
Wasserstein distance between human and LM opinion distributions

OpinionQA: Measuring Opinion Alignment in Language Models

- ▶ **Build:** OpinionQA dataset with 1,500 questions across various topics
 - ◉ Sourced from Pew's American Trends Panels
 - ◉ Includes opinion distributions for the US population and 63 demographic groups
- ▶ **Evaluate:** Tested 9 language models from OpenAI and AI21 on OpinionQA
 - ◉ Measured **opinion alignment** to human responses along three key axes

Axis 1: Representativeness

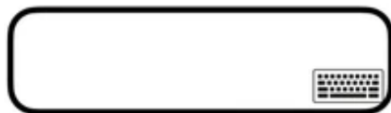


Humans		AI21 Labs			OpenAI					
Avg	Worst	j1-grande	j1-jumbo	j1-grande-v2-beta	ada	davinci	text-ada-001	text-davinci-001	text-davinci-002	text-davinci-003
0.949	0.865	0.813	0.816	0.804	0.824	0.791	0.707	0.714	0.763	0.700

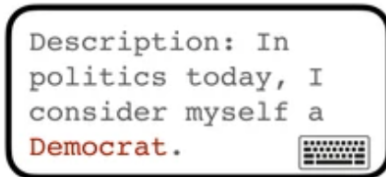
- ▶ LM-US populace opinion alignment $\approx$ Democrats-Republicans on climate change and gets worse for newer models
- ▶ Shift between "base" LMs and human-feedback (HF) trained ones: towards **more educated, wealthy and liberal**

Axis 2: Steerability

"DEFAULT"

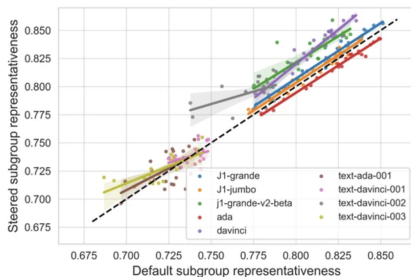


"STEERED"



Axis 2: Steerability

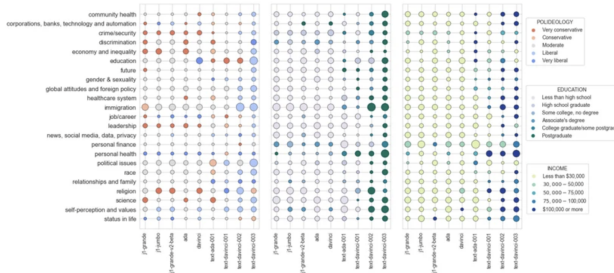
Defn: Can we **steer** LM opinions towards those of **specific** groups via prompting?



→ LMs are somewhat steerable **but** performance disparities between groups still remain

Axis 3: Consistency

Defn: How consistently do LM lean towards those of **specific group across topics**?



→ LMs reflect a patchwork of viewpoints

→ Even “liberal” LMs tend to lean conservative on topics such as religion

Toward Pluralistic Alignment

- ▶ Current alignment pipelines (SFT / RLHF) often push models toward the **center of the distribution**
 - ⊙ Reflecting the preferences of specific annotators or demographics
- ▶ But societies are not homogeneous
 - ⊙ People differ in culture, values, and priorities
- ▶ Emerging research direction: *Pluralistic Alignment*
 - ⊙ How can models represent **multiple perspectives**?
 - ⊙ How can users **steer models toward different value systems**?

Open Question

How can we build AI systems that respect **human plurality** rather than converging to a single global viewpoint?

Where are we with LLMs?

Some Questions to Think About

Where Are We With LLMs?

Open Research Questions

► Reasoning

- ◉ Are LLMs **truly reasoning**, or still relying on pattern matching?

► Scale vs Efficiency

- ◉ Is scale all we need, or can **smaller models** achieve similar capabilities?

► Reliability

- ◉ How do we detect and prevent **hallucinations**?

► Interpretability and Control

- ◉ How can we better **interpret** and **control** LLM behavior?

► Alignment

- ◉ How can we build AI systems that represent **human plurality**?



That's all
for the course!

Keep asking good questions about
LLMs, NLP, and AI!